

AI

AI Check System

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AI Assignment Report

File: assign_8_6a49c421d8fec9.09383636.pdf

User ID: 8

User Name: " . esc(Atikah Zanal) . "

Date Processed: July 5, 2026 at 4:44 AM

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AI CHECKED / COMPLETED

⚙️ **AI Analysis Results**

2

AI Probability Score (0-100)

Risk Level: Low

39378

Words

728

Sentences

1

Paragraphs

count(Array)

Keywords

💬 **AI Feedback**

The assignment appears to be predominantly human-written content.

Structural patterns, vocabulary usage, and sentence variability are consistent with natural writing.

Content Statistics: 39378 words, 728 sentences, 1 paragraphs.

2 academic keywords identified and highlighted in the report.

🎨 **Detected Keywords**

data

review



Original Assignment Content

Keywords highlighted in yellow . File: assign_8_6a49c421d8fec9.09383636.pdf

(IJACS) International Journal of Advanced Computer Science and Applications Vol. 14, No. 9, 2023 366 | Page www.ijacsa.thesai.org Exploring the Challenges and Impacts of Artificial Intelligence Implementation in Project Management: A Systematic Literature Review

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Abstract This paper presents a systematic literature review (SLR) investigating the challenges and impacts of implementing artificial intelligence (AI) in project management, specifically mapping them into the process groups defined in the Project Management Body of Knowledge (PMBOK). The study aims to contribute to the understanding of integrating AI in project management and provides insights into the challenges and impacts within each process group. The SLR methodology was applied, and a total of 34 scientific articles were analyzed. The results and analysis reveal the specific challenges and impacts within each process group. In the Initiating Process Group, AI tools and analysis techniques address challenges in risk assessment, cost prediction, and decision-making. The Planning process group benefits from various tools and methodologies that improve risk assessment, project selection, cost estimation, resource allocation, and decision-making. The Execution process group emphasizes the importance of advanced tools and techniques in enhancing productivity, resource utilization, cost reduction, and decision-making. The Monitoring and Controlling process group demonstrates the potential of advanced tools in achieving efficiency, cost reduction, improved quality, and informed decision-making. Lastly, the Closing process group emphasizes the importance of utilizing advanced tools to minimize waste, optimize resource utilization, reduce costs, improve quality, and project closure success. Overall, this research provides valuable insights and strategies for organizations seeking to implement AI in project management, thereby enhancing the potential for success within the PMBOK Process Group.

Keywords Artificial intelligence; project management; PMBOK process groups; challenge; impact.

1. INTRODUCTION Industry 5.0 is an evolution of Industry 4.0, focusing on a more human-centric approach while leveraging advanced technologies like Artificial Intelligence (AI) and big data. It aims to create a sustainable and resilient industry by combining technological advancement with human needs [1], [2]. Project management has experienced a paradigm shift because of the quick development of AI technology. The way projects are planned, carried out, and controlled may be completely transformed using AI [3], [4]. The incorporation of AI into project management techniques, however, comes with a unique set of difficulties and has a big influence. Organizations need to address these challenges to fully harness the potential of AI in project management [5], [6]. Therefore, it is essential to investigate and understand the challenges faced and impacts observed during the implementation of AI in project management [6]. Given the transformative impact of AI, there is an increasing signifi-

ican ce in s tu dy in g th e p ro ce ss o f A I a d op ti on . Nu me ro us s tu di es h av e b ee n co nd u ct ed to ex pl o re th e d ri vi ng fa ct o rs , ba r ri e rs , ch al le ng es , an d o ve r al l p er fo r ma nc e im pa ct as so ci at ed w it h A I im pl e m en ta ti on in o r ga ni za ti on s [6] , [7] . Th e ac cu ra cy an d a p pr o p ri at e ne ss o f da ta p er ta in in g to p ro j ec t ma na ge m en t to ol s ar e cr u ci al . In it ia l A I to ol s fo r p ro j ec t ma na ge m en t h ea vi ly d ep en d on in di vi du al s to ac cu ra te ly in pu t da ta , ti me ly up da te to ol s , an d ma ke ne ce ss a ry co r re ct io ns [8] . Th e in te gr a ti on o f A r ti fi ci al In te ll ig en ce (A I) in to bu si ne ss o pe ra ti on s h as gi ve n ri se to th e em er ge nc e o f in te ll ig en t en vi ro nm en ts , in cl u di ng ad va nc e d mo ni to ri ng sy st em s fo r p ro j ec t ma na ge m en t . Si mi la r ly , A I h as b ee n em br ac ed in th e fi el d o f p ro j ec t ma na ge m en t , of fe ri ng p ro m is in g p ro sp ec ts fo r th e fu tu re o f p ro j ec t ma na ge m en t ac ti vi ti es [3] , [9] . Mo re o ve r , A I h as th e ca pa bi li ty to mo ni to r p ro j ec t s ta tu s an d ma ke ad ju st m en ts wh en re qu ir ed . In te gr a ti ng A I in to p ro j ec t ma na ge m en t al lo ws fo r mi ni mi al hu ma n in te r ve n ti on by u ti li zi ng in te ll ig en t ma ch in es an d la rg e da ta se ts to au to ma te de ci si on - ma ki ng an d ta sk ma na ge m en t . Th is au to ma ti on en ab le s A I to p la y a gu id in g ro le in p ro j ec ts , au to ma ti ng ta sk s an d ai di ng in de ci si on - ma ki ng p ro ce ss es [6] , [10] , [11] . Th is pa pe r ai ms to co nd u ct a sy st em a ti c li te ra tu re re vi ew (SL R) to in ve st ig a te th e ch al le ng es fa ce d an d im pa ct s ob se r ve d du ri ng th e im pl e m en ta ti on o f A I in p ro j ec t ma na ge m en t , sp e ci fi ca lly ma p pi ng th em in to th e p ro ce ss g ro up s de fi ne d in th e PM B OK [12] , [13] . Th e re se ar ch qu es ti on s de ri ve d fr om th is ob j ec ti ve ar e as fo ll ow s : R Q1 : Wh at ar e th e ch al le ng es fa ce d an d im pa ct s ob se r ve d du ri ng th e im pl e m en ta ti on o f A I in p ro j ec t ma na ge m en t ? R Q2 : Ho w do es th e im pl e m en ta ti on o f A I in p ro j ec t ma na ge m en t in co r po ra te th e m ap pi ng o f ch al le ng es an d im pa ct s in to th e p ro ce ss g ro up s de fi ne d in th e PM B OK ? By co nd u c ti ng a co mp re he ns iv e re vi ew o f ex is ti ng li te ra tu re , th e fi nd in gs wi ll p ro vi de in si gh ts in to th e ch al le ng es an d im pa ct s sp e ci fi c to ea ch p ro ce ss g ro up s , h el pi ng o r ga ni za ti on s ef fe ct iv e ly ad dr es s th es e ch al le ng es an d le ve ra ge th e po te n ti al be ne fi ts o f A I in p ro j ec t ma na ge m en t . Th is pa pe r fo ll ow s a sy st em a ti c p ro ce ss , co m m en ci ng w it h a Li te ra tu re Re vi ew se ct io n th at ev al ua te s ex is ti ng s tu di es o n A I im pl e m en ta ti on an d p ro ce ss g ro up s in p ro j ec t ma na ge m en t . (IJACS A) In te r na ti on al Jo ur na l o f Ad va nc e d Co m pu te r Sc ie nc e an d Ap pl i ca ti on s Vo l . 14 , No . 9 , 2023 367 | Pa ge www . ij ac s a . th es ai . o r g Th en th e Me th o do lo gy se ct io n co ns is ts o f th re e ph as es : p la n ni ng th e Sy st em a ti c Li te ra tu re Re vi ew (SL R) , ex e cu ti ng th e SL R , an d p re se n ti ng th e SL R fi nd in gs . W it h in th e Re su lts an d A na ly si s se ct io n , tw o su b se ct io ns ar e de li ne at ed . Th e fi rs t su b se ct io n m ap s A I to ol s , im pl e m en ta ti on ch al le ng es , an d im pa ct s ba se d on p ri o r re se ar ch , wh il e th e se co nd su b se ct io n al ig ns A I to ol s , ch al le ng es , an d im pa ct s w it h PM B OK P ro ce ss G ro up s (in it ia ti ng , p la n ni ng , ex e cu ti on , mo ni to ri ng an d co n tr ol li ng , cl os in g) . La st ly , th e pa pe r co n cl u de s by su m ma ri zi ng ke y fi nd in gs , di sc us si ng th ei r im pl i ca ti on s fo r p ro j ec t ma na ge m en t , an d su gg es ti ng di re ct io ns fo r fu tu re re se ar ch . II. LI TE RA TU RE RE VI EW A . A r ti fi ci al In te ll ig en ce an d Da ta Mi ni ng A r ti fi ci al in te ll ig en ce (A I) is a fi el d w it h in co m pu te r sc ie nc e th at ai ms to de ve lo p in te ll ig en t ma ch in es fo r th e be ne fi t o f hu ma ns . Ho we ve r , th er e is on go in g de ba te am on g re se ar ch er s re ga r di ng th e de fi ni ti on o f A I du e to its ev ol vi ng na tu re . Va ri ou s de fi ni ti on s ex is t in th e li te ra tu re , re fle c ti ng re se ar ch er s ' sp e ci a li za ti on s an d in te re st s , al l st ri vi ng to ex pl ai n th e co nc ep t o f A I an d p ro vi de a wi de ra ng e o f te ch no lo gi es th at en ha nc e p er fo r ma nc e an d in te ra ct io n w it h in o r ga ni za ti on s [3] , [14] . A I h as tra di ti on a lly b ee n ap pr o ac he d fr om fo ur p er sp ec ti ve s : th in ki ng an d ac ti ng , wh ich en co m pa ss th ou gh t p ro ce ss es , re as on in g , an d be ha vi o r . Th es e p er sp ec ti ve s ca n b e fu r th er ca te go ri ze d in to a hu ma n - ce n te r ed ap pr o ac h ba se d on hu ma n be ha vi o r ob se r va ti on s , an d a ra ti on a li st ap pr o ac h co m bi n

in g m a t h e m a t i c s a n d e n g i n e e r i n g c o n c e p t s [1] . **Data** m i n i n g i s a b r a n c h o f M a c h i n e L e a r n i n g a n d A r t i f i c i a l I n t e l l i g e n c e t h a t i n v o l v e s a n a l y z i n g a d a t a s e t u s i n g a l g o r i t h m s t o u n c o v e r p a t t e r n s a n d r e l a t i o n s h i p s . I t a l l o w s u s e r s t o a n a l y z e d a t a f r o m m u l t i p l e p e r s p e c t i v e s , c a t e g o r i z e i n f o r m a t i o n , a n d d r a w c o n c l u s i o n s . T h e p r o c e s s i n v o l v e s s e a r c h i n g f o r c o r r e l a t i o n s b e t w e e n d i f f e r e n t f i e l d s i n a d a t a b a s e t o e x t r a c t v a l u a b l e i n s i g h t s a n d i n f o r m a t i o n [1 5] [1 7] . M a c h i n e l e a r n i n g e n a b l e s t h e c a t e g o r i z a t i o n o f d a t a p r o c e s s i n g m e t h o d s i n d a t a m i n i n g i n t o d i s t i n c t c a t e g o r i e s s u c h a s c l a s s i f i c a t i o n , r e g r e s s i o n a n a l y s i s , a s s o c i a t i o n r u l e s , a n d c l u s t e r i n g . E a c h o f t h e s e m i n i n g m e t h o d s c a n b e e x e c u t e d u s i n g v a r i o u s m a c h i n e l e a r n i n g t e c h n i q u e s [1 8] .

B. Process Groups of Project Management

Project management procedures are organized into logical groups of inputs, tools and techniques, and outputs that are tailored to the needs of the organization, the project, and the stakeholders. Instead of being interchangeable with project phases, these process groups work together during each stage of the project's life cycle. In order to maintain flexibility and adaptation throughout the project, the number of iterations and interactions across processes can change based on the demands of the project [1 2] , [1 3] . As shown in Fig. 1, participation with the remaining Process Groups is required to fully realize the collaborative aspect of project management. Projects that adopt a process-based approach can be structured into five groupings based on different processes: 1) The initiating phase of a project involves defining and obtaining authorization for a new project or phase [1 2] . This phase includes creating a project charter and implementing a formalized project initiation process to support project management decisions and ensure project success [1 9] . Fig. 1. Project management process groups [1 3] . 2) The planning phase of a project involves establishing the project's scope, refining objectives, and determining the necessary actions to achieve those objectives [1 2] . This phase encompasses defining the course of action, making decisions, and formalizing the project's direction. Planning is a crucial step in project management, involving the conscious determination of actions to accomplish goals, and it is typically conducted after business planning and before project execution in fields such as construction [2 0] . 3) The execution phase involves completing the tasks listed in the project management plan in order to comply with the project's criteria [1 2] . It encompasses the actual performance of project tasks and activities as outlined in the plan to fulfill the project's objectives [2 0] . 4) The monitoring and controlling phase of project management involves tracking progress, reviewing performance, and making necessary changes to ensure project success [1 2] . Studies have emphasized the importance of timely control information and the ability to handle unexpected crises as critical factors in effective monitoring and control [2 0] . 5) The closing phase of project management involves formally completing all activities and closing the project, phase, or contract [1 2] . Research conducted by Ampoah [2 1] focused on project failure/success factors in Ghana's agriculture, banking, and construction sectors. The study identified project management tools, techniques, and methods used by project managers and recommended that companies in Ghana focus on improving their project management activities for better outcomes.

C. Challenges and Impact of AI Implementation in Project Management

Over time, project management has undergone significant changes and has become increasingly valuable to organizations. The collaboration between humans and machines in the face of major technological advancements presents significant opportunities for both businesses and individuals. One such advancement is Artificial Intelligence (

AI), which is expected to greatly influence the role of project managers [6]. AI, despite its complexities, has the potential to (IJACS A) International Journal of Advanced Computer Science and Applications Vol. 14, No. 9, 2023 368 | Page www.ijacs.a.thesai.org increase productivity and reduce errors in various fields, including software development projects. By providing insights into probable outcomes and removing extraneous information, AI improves project management and enables a focus on relevant facts [22]. The challenges of implementing AI in project management include barriers such as limited data availability, high costs of operation and equipment, and potential unemployment as AI replaces human workers. The successful adoption of AI requires technical staff with specialized skills and experience, along with a clear understanding of system integration and interoperability challenges [3]. While AI offers benefits like cost-effectiveness and reliability, it also raises issues of uncertainty and highlights both the positive and negative aspects of its adoption [23]. The adoption of AI in project management enhances decision-making quality by providing insights and support for potential outcomes. AI systems streamline information by eliminating redundancy, and auto-scheduling improves the robustness of project planning. Planning tools powered by AI, including hybrid computer systems, facilitate project control and objective configuration [8]. While contemporary management strategies like continuous alignment and agile approaches help overcome uncertainties, predictive analytics and machine learning contribute to better project outcomes in areas like KPIs, resource management, and estimation [3], [24].

III. METHODOLOGY

This section explains the literature review process using the Systematic Literature Review (SLR) method. This method is used to evaluate and interpret literature related to the research questions, topic areas, or desired phenomena [25]. The goal of Systematic Literature Review is to generate accurate evaluations of the research topic using reliable, meticulous, and auditable methodologies. It is divided into three parts based on the Kitchenham [25] and references [26], [27]: Planning, Implementation, and Reporting of SLR stages. A. Planning the SLR In this stage, keyword generation is conducted based on the main keywords and synonyms obtained at the beginning of the research, namely Artificial Intelligence and Project Management. Subsequently, these keywords are combined using "AND" to connect the main keywords and "OR" for each synonym or simpler word. The keywords and their synonyms are required as follows: Artificial Intelligence synonym These keywords and their synonyms are combined to form the search keywords as follows: OR OR Then the search is conducted on several reputable literature and journal websites, based on title, abstract, or keywords. The databases are IEEE Xplore, Springer Link, Emerald Insight, SAGE Journals, Science Direct, Scopus, and ACM Digital Library. After determining which database to use for the search, the researcher proceeds with the study selection process. This is intended to identify key studies with a research as direct evidence related to the previously obtained keywords [25]. The criteria for the study selection process in this research can be seen in Table I.

TABLE I. CRITERIA FOR STUDY SELECTION

Inclusion Criteria IC1 Publications written in English IC2 Publications starting from 2018 until 2023 IC3 Publications that truly focus on the keywords: artificial intelligence/data mining/machine learning and project management IC4 Publications that increasingly focus on the root problem, specifically discussing the challenges and impact of AI/DM/ML on project management

Exclusion Criteria EC1 Research not written in English EC2 Research that discusses

stopicsotherthanAI/DM/MLandprojectmanagementEC3Researchpublishedbefore2018B.ImplementationofSLRInthesecondphaseoftheSystematicLiteratureReview(SLR),thefocusisonconductinganextensivesearchandselectingrelevantliterature.Thisinvolvesidentifyingpertinentstudies,extractingrelevantinformation,andsynthesizingthefindingstoobtainacomprehensiveunderstandingoftheresearchtopic.Fig.2representstheflowdiagramillustratingtheselectionprocessconductedfromtheinitialdatacollectioninvariousdatabasesusingtheinclusionandexclusioncriteriaasspecifiedinTableI.Fig.2.FlowdiagramofSLRimplementation.(IJACSA)InternationalJournalofAdvancedComputerScienceandApplicationsVol.14, No.9, 2023 369 | Page www.ijacsa.thesai.org Thetotalnumberofarticles thatwillundergofurtheranalysisisrelatedtothechallengesandimpacts onprojectmanagementamountsto34articles,obtainedfromvarioushighindexjournalsandproceedingsbasedonScimago.C.ReportingtheSLRThefinalstepoftheSystematicLiteratureReview(SLR)isreportingtheresults.TableIIpresentsasummaryofallrelevantarticletitlesalignedwiththeresearchobjectives,includinginformationontheyearandreferencecitationcoding.TABLERELEVANTARTICLES

No	Title	Year	Index Code
1	ABIM-dataminingintegrateddigitaltwinframeworkforadvancedprojectmanagement	2021 Q1	[28]
2	AMachineLearningStudytoEnhanceProjectCostForecasting	2022 Q3	[29]
3	Anewhybridahpanddempstershafertheoryofevidencemethodforprojectriskassessmentproblem	2021 Q2	[30]
4	Activityclassificationusingaccelerometersandmachinelearningforcomplexconstructionworkeractivities	2021 Q1	[31]
5	AnApproachBasedonBayesianNetworkforImprovingProjectManagementMaturity:AnApplicationtoReduceCostOverrunRisksinEngineeringProjects	2020 Q1	[32]
6	ApplicationofDataMiningTechnologyinFieldVerificationofProjectCost	2021 Q4	[33]
7	Applicationofleantechniques,enterpriseresourceplanningandartificialintelligenceinconstructionprojectmanagement	2019 Q4	[34]
8	AutomatedprogressmonitoringofconstructionprojectsusingMachinelearningandimageprocessingapproach	2022 Q2	[35]
9	Combinedmachine-learningandEDMtomonitorandpredictacomplexprojectwithaGERT-typenetwork:Amulti-pointperspective	2023 Q1	[36]
10	Comprehensiveprojectmanagementframeworkusingmachinelearning	2019 Q4	[37]
11	DataonFieldCanalsImprovementProjectsforCostPredictionUsingArtificialIntelligence	2020 Q2	[38]
12	Data-drivenprojectbuffersizingincriticalchains	2022 Q1	[11]
13	Decision support system for final year project management	2019 Procd	[39]
14	Developmentandcomparativeofanewmeta-ensemblemachinelearningmodelinpredictingconstructionlaborproductivity	2022 Q1	[40]
15	DevOpsprojectmanagementtoolsforsprintplanning,estimationandexecutionmaturity	2020 Q2	[41]
16	EmpiricallyExploringtheCause-EffectRelationshipsofAICharacteristics,ProjectManagementChallenges,andOrganizationalChange	2021 Procd	[5]
17	Estimatingproductionandwarrantycostattheearlystageofanewproductdevelopmentproject	2021 Q3	[42]
18	EstimationofRiskContingencyBudgetinProjectsusingMachineLearning	2022 Q3	[43]
19	Explainablemachinelearningforprojectmanagementcontrol	2023 Q1	[44]
20	Forecastingtheschedulingissuesinengineeringprojectmanagement:Applicationsofdeeplearningmodels	2021 Q1	[9]
21	InformationTechnology(IT)GovernanceFrameworkwithArtificialNeuralNetworkandBalanceScorecardtoImprovethethe		

SuccessRateofSoftwareProjects2022Procd[45]22Intelligentpurchasing: Howartificialintelligencecanredefinethepurchasingfunction2021Q1[46]23Machinelearninginprojectanalytics:adata-drivenframeworkandcase study2022Q1[47]24ProjectengineeringmanagementevaluationbasedonGABPneuralnetworkandartificialintelligence2023Q2[48]25Projectmanagement:openingsfordisruptionfromAlandadvancedanalytics2021Q1[49]26ProposalofaframeworkandintegrationofartificialintelligencetosucceedITprojectplanning2019Q4[50]27RecommendationofProjectManagementPractices: AContributiontoHybridModels2022Q1[51]28Safetyriskfactorscomprehensiveanalysisforconstructionproject: Combinedcascadingeffectandmachinelearningapproach2021Q1[52]29Symbioticorganisms search-optimizeddeeplearningtechniqueformappingconstructioncashflowconsideringcomplexityofproject2020Q1[53]30Theeffectivenessofprojectmanagementconstructionwithdataminingandblockchainconsensus2021Q1[54]31Theimpactofentrepreneurshiporientationonprojectperformance: A machinelearningapproach2020Q1[55]32Thevalueofdatafromconstructionprojectsitemeetingminutesinpredictingprojectduration2022Q2[56]33UsinganArtificialNeuralNetworkforImprovingthePredictionofProjectDuration2022Q2[57]34VisualSystemDevelopmentforConstructionProjectManagementbyUsingMachineLearningAlgorithm2022Q2[58](IJACS A) International Journal of Advanced Computer Science and Applications Vol. 14, No. 9, 2023 370 | Page www.ijacsa.thesai.org Furthermore, the distribution of the scientific articles based on their publication years is presented in Fig. 3. Fig. 3. Distribution of study per year. IV. RESULTS AND ANALYSIS In accordance with the objective of this research, a thorough mapping and analysis of the literature obtained will be conducted using the Systematic Literature Review (SLR) method described in the previous chapter. The outcomes obtained from this process will be divided into the mapping of AI tools used, challenges, and the impact of AI implementation on each article, as well as their summaries. A Mapping the AI Tools, Implementation Challenges and Impacts based on Related Studies Table III presents the further analysis results of the articles obtained in the previous SLR process, where the analysis is conducted regarding the AI tools used in the referenced studies, as well as the challenges and impacts of implementation in those studies. These AI tools can include algorithms, software, models, or others. TABLE III. MAPPING THE AI TOOLS, CHALLENGES, AND IMPACT OF THE AI IMPLEMENTATION Ref AI Tools Challenge Impact [28] BIM, IoT, DM techniques, process mining, timeseries analysis, fuzzymineralgorithm, ARIMAX model Real-time IoT data mapping and complexity management in large-scale projects. A lenable early bottleneck detection, future workload prediction, and proactive decision-making for enhanced efficiency and adaptability. [29] Moving averages, schedule and cost performance factors Nonlinear budget acquisition and cost growth patterns not accurately captured by traditional index-based models. AI provides reliable and accurate cost estimates, improving project cost forecasting and maintaining positive stakeholder relationships. [30] Hybrid PCA-agglomerative unsupervised ML algorithm, Delphi method, statistical analysis, Taguchi Method Uncertainty and variability of risk factors not accurately addressed by traditional decision-making methods. AI quantifies risk factors for more informed project selection and impr

oved decision-making, potentially enhancing project success rates. [31] Machine learning algorithms (e.g., K-Nearest Neighbors, Gradient Boosting), wearable accelerometers, feature selection techniques Dealing with realistic scenarios, bias minimization, imbalanced datasets, and optimizing classifier performance with activity transitions. AI enhances project decision-making, resource management, worker performance, and safety in construction, contributing to automation and process-level knowledge. [32] Bayesian networks Integrating PMM models with Bayesian networks, formalizing expert knowledge, and addressing sector-specific factors and project drifts. AI-based PMM diagnoses and predicts project performance failures, identifies drifts, and suggests corrective actions for comprehensive project success analysis. [33] Statistical data mining technology (not specified) Developing a comprehensive organizational structure, implementing advanced management practices, improving cost communication, and addressing cost verification shortcomings. AI introduces a data-driven approach, improving visualization, resource management, data collection, and cost control in project management. [34] Voice command, optical character recognition, LiDAR sensors Integrating multiple systems, ensuring compatibility, overcoming resistance to change, and addressing complexity and uncertainty in construction projects. AI minimizes waste, optimizes resource utilization, reduces costs, improves delivery time and quality, and enhances productivity in the construction industry. [35] Deep learning models, supervised CNN classifiers, image processing techniques Reluctance, data standardization, and accuracy of automated tracking Supervised CNN classifier improves activity recognition, enhancing supervision and progress tracking in construction projects. [36] Monte Carlo simulation, machine learning regression algorithms Analyzing complex networks and selecting suitable algorithms AI improves project monitoring, deviation identification, and completion time estimation, enhancing project performance and decision-making. [37] Artificial Intelligence (AI), machine learning models, online solutions (not specified) Model selection, training, integration, data accuracy, and user interface development. AI improves planning, decision-making, automation, and holistic solutions for project management. [38] Machine learning and AI techniques (Regression, Fuzzy, ANN, SVM, and others) Dataset quality, algorithm selection, and interpretation/validation. Understanding factors influencing project outcomes improves planning and resource allocation in sustainable projects. [11] Full-factorial design of experiments, Monte Carlo simulation, regression techniques Designing experiments, simulations, and ensuring model validity, AI-based buffersizing improves project planning and timeliness compared to traditional methods. [39] Naïve Bayes algorithm, JSP technology Database organization, accuracy of Naïve Bayes, and technical issues. AI platform for matching students with project topics and supervisors improves the quality and efficiency of final year research projects. [40] Ensemble machine learning algorithms Preprocessing, algorithm selection, converting models, and case studies. Accurate labor productivity prediction and influential factors guidance improve construction project management and overall productivity. (IJACS) International Journal of

Advanced Computer Science and Applications Vol. 14, No. 9, 2023 371 | Page www.ijacsa.thesai.org [41] Real-time project data analysis, sprint estimation techniques, sentiment analysis Real-time data analysis, accurate predictions, consensus, and sentiment analysis. AI tools streamline project timelines, increase productivity, and improve decision-making in a DevOps environment. [5] AI techniques, machine learning algorithms, data analysis methods Addressing AI's characteristics and implications in organizational settings. Understanding the relationships between AI characteristics, project management practices, and organizational change improves AI implementation and decision-making. [42] Artificial neural network, linear regression techniques Defining variables, acquiring data, and simulating cost variants. Systematic consideration of cost factors and constraints improves cost estimation and optimization in new product development projects. [43] Machine learning algorithms, Monte Carlo simulation Defining risks, selecting algorithms, and ensuring accuracy Integration of machine learning techniques enhances Contingency Budget estimation accuracy and risk management in projects. [44] Monte Carlo simulation, statistical/machine learning models, explainable machine learning with SHAP Integrating simulation and explainability, selecting models, and handling complexity. Integration of Monte Carlo simulation, machine learning, and explainable machine learning improves project control and decision-making in uncertain environments. [9] LSTM and GRU models Forecasting model selection, data quality, and interpretation. LSTM and GRU models enable effective scheduling, resource allocation, and decision-making, improving overall project efficiency. [45] Artificial Neural Network (ANN) method, Balanced Score Card framework Top management commitment, solution finding, and data availability. AI integration in software project management enables informed decision-making and better outcomes, leading to the development of new project life-cycle solutions. [46] Automated systems for purchasing, and decision support Resistance, integration, data quality, and training. AI systems redefine purchasing roles, improved decision-making, supplier relations, and interdepartmental collaboration, highlighting the potential of AI capabilities in supplier management. [47] Machine learning algorithms (e.g., support vector machine, logistic regression, neural networks) using Python's Scikit-learn package Limited evaluation, feature selection, and data imbalance. AI-driven data analysis improves decision-making and understanding in construction project management. [48] BP neural network, genetic algorithms Data availability, neural network limitations, and minimizing subjectivity. AI-based system optimizes time, personnel, and resource utilization, improving engineering project management efficiency. [49] Software tools with AI and analytics features Project managers adapting to software features AI and analytic tools enhance project management with increased support, automation, and adaptive practices, emphasizing stakeholder relations and risk management. [50] Knowledgebase, questioning system Data acquisition, correctness, and risk management. AI-based solution improves accuracy and efficiency of IT project planning, reducing failure rates and enabling cost savings. [51] Cluster analysis, association rule technique Data collection, accuracy, and validation

ion. Guidance for selecting practices to enhance project management agility and effectiveness, suggesting avenues for further research in customizing hybrid models. [52] Machine learning, safety risk factor mining, AON networks Abundant data, algorithm optimization, and quantifying risk impact. AI improves safety risk analysis in construction projects, enhancing risk management, with further research needed for data independence and cost considerations. [53] Symbiotic organisms search (SOS) algorithm, LSTM, neural networks Cashflow, model optimization, and data availability. AI model accurately forecasts and controls cashflow in construction projects, improving cost management, decision-making, and resource allocation. [54] AI middle office, AI algorithms, blockchain technology, BIM Data availability, integration, and consensus building. Integration of AI middle office, BC technology, and BIM technology enhance trust, transparency, accuracy, security, and project management efficiency. [55] Machine learning algorithms (e.g., lasso, ridge, support vector machines, neural networks, random forest) AI algorithm selection, performance metrics, self-assessment, and generalizability. AI techniques provide insights into operations and project management research, highlighting the potential of predictive analytics and entrepreneurial orientation for projects success. [56] Data mining algorithms, random forest Meeting minute data extraction, time-consuming capture, and data quality. AI enables accurate project duration prediction, facilitating timely actions and improving project planning, leadership, and governance. [57] Artificial neural networks, genetic algorithms Diverse datasets, adaptation to different organizations, and validation. AI accurately predicts project duration for different organizations, adapting to various methods and datasets, enhancing project management decision-making. [58] Graph neural network, deep learning algorithms Complex building data, accuracy of traditional algorithms, and limited early-stage. Integration of complex data and machine learning algorithms improves building management efficiency, information access, communication, energy conservation, and safety. Artificial intelligence tools in project management based on the related studies mentioned include BIM, IoT, DM techniques, process mining, time series analysis, fuzzy miner algorithm, ARIMA X model, moving averages, schedule and cost performance factors, hybrid PCA-agglomerative unsupervised ML algorithm, Delphi method, statistical analysis, Taguchi Method, machine learning algorithms (such as K-Nearest Neighbors and Gradient Boosting), wearable accelerometers, feature selection techniques, Bayesian networks, voice command, optical character recognition, LiDAR sensors, deep learning models, supervised CNN classifiers, image processing techniques, Monte Carlo simulation, artificial neural networks, full-factorial design of experiments, Naïve Bayes algorithm, ensemble machine learning algorithms, real-time project data analysis, sprint estimation techniques, sentiment analysis, software tools with AI and analytics features, knowledge base, questioning system, cluster analysis, association rule technique, safety risk factor mining, symbiotic organisms search (SOS) algorithm, LSTM, (IJACS A) International Journal of Advanced Computer Science and Applications Vol. 14, No. 9, 2023 372 | Page www.ijacs.a.thesai.org AI middle office, lasso and ridge regression, support vector machines, random forest, data mining algorithms, genetic algorithms, and graph neural network. From a project management

elements stand point, AI has the capability to replicate human cognitive functions such as decision-making and problem-solving [3]. These tools bring advanced capabilities to project management, enabling tasks such as data analysis, risk assessment, decision-making, performance monitoring, and optimization [42], [49]. They leverage AI, machine learning, and statistical techniques to improve project planning, resource allocation, cost estimation, risk mitigation, and overall project outcomes [43], [52]. These tools enhance efficiency, accuracy, and insights in project management processes, ultimately leading to improved project success rates and delivery [50], [54].

B. Mapping AI Tools, Challenges, and Impacts of Articles based on PMBOK Process Group

Next is to map the process groups of PMBOK, which according to the literature are divided into 5, namely Initiating, Planning, Execution, Monitoring and Controlling, and Closing [12]. This mapping aims to integrate project management processes and product-oriented processes, ensuring proper coordination and alignment. The function of mapping the process group is to achieve this integration throughout the project life cycle by tailoring the application of processes to meet the project's specific requirements and actively managing process interactions and trade-offs to meet stakeholder needs [13]. The mapping results are represented in Table IV. Based on the mapping results of the reference articles to the Process Groups in PMBOK, the next step involves reviewing the summaries of AI Tools used in each stage of the Process Group. Additionally, the challenges and their respective impacts on each process will be explored based on the conducted literature study.

TABLE IV. MAPPING RELATED STUDIES TO PROCESS GROUP

No	Process Group	PMBOK Reference	Freq
1	Initiating	[30], [38]	2
2	Planning	[5], [11], [29], [30], [32], [34], [37], [38], [41]	4
3	Execution	[34], [37], [39]	4
4	Monitoring and Controlling	[9], [28], [31], [33]	3
5	Closing	[34], [49]	2

1) Initiating process group: In the Initiating process group of the PMBOK, both papers discuss the application of AI tools and analysis techniques to address challenges in risk management and cost prediction in projects. The first paper [30] utilized a hybrid AHP and Dempster-Shafer methods along with various AI tools such as machine learning algorithms, Delphi method, statistical analysis, and Taguchi Method to tackle the challenge of uncertainty and variability in risk factors. These AI tools had a significant impact by offering a more reliable approach to project selection through the consideration and quantification of risk factors, enabling informed decision-making and potentially improving project success rates. Similar to the first article, the second paper [38] used Machine Learning and AI methods to estimate and forecast the cost and length of field canal development projects. These methods included regression, fuzzy logic, artificial neural networks, support vector machines, and others. Despite challenges related to data quality, algorithm selection, and interpretation/validation, the application of these techniques positively influenced the understanding and identification of factors influencing project outcomes, leading to improved planning and resource allocation in sustainable projects. In summary, within the Initiating process group, the integration of AI tools and analysis techniques in project management addresses challenges in risk assessment, cost prediction, and decision-making. These advancements provide more reliable approaches for project selection, consideration and quantification of risk factors, and improving the overall success rates of projects. Additionally, the utilization of Machine Learning and AI techniques enhances the understanding of project outcomes and aids in better planning and resource allocation.

ocation. However, challenges such as uncertainty, variability, and data set quality need to be carefully addressed to ensure accurate predictions and interpretations in project management. 2) Planning process group: This comprehensive analysis summarizes the key findings and contributions of 19 scientific papers related to project management. These papers utilize various tools and techniques such as machine learning, artificial intelligence (AI), data mining, and statistical analysis to address challenges and enhance the process of group planning within the PMBOK framework. By leveraging these advanced tools, project managers can enhance decision-making, risk assessment, resource allocation, and project planning. The application of these tools and methodologies in project management has significant impacts. By incorporating hybrid models, advanced algorithms, and statistical techniques, projects can benefit from improved risk assessment, more reliable project selection, accurate cost estimation, optimized resource utilization, and enhanced decision-making [29], [30], [38]. These advancements contribute to higher project success rates, improved project outcomes, and efficient resource allocation. Furthermore, the integration of machine learning, artificial intelligence, and data analytics in project management processes enhances real-time data analysis, sentiment analysis, and decision support. This results in streamlined project timelines, increased productivity, and improved project management within a DevOps environment [41]. Additionally, the adoption of AI and advanced analytics tools in project management opens opportunities for disruption, automation, adaptive practices, and better stakeholder relations and risk management [32]. The construction industry also benefits from the application of AI, machine learning, and lean techniques. These (IJACS A) International Journal of Advanced Computer Science and Applications Vol. 14, No. 9, 2023 373 | Page www.ijacs.a.thesai.org technologies enable the optimization of resource utilization, reduction of project costs, improvement of project delivery time and quality, and enhancement of overall productivity and efficiency [34], [52]. Moreover, the integration of AI in the South African construction project management industry contributes to job security, accident reduction, automation of high-risk tasks, and the creation of new job opportunities, thereby fostering regional development [59]. In summary, the 19 scientific papers examined show a wide range of tools and methodologies that address various challenges in project management. The application of these tools has a profound impact on risk assessment, project selection, cost estimation, resource allocation, decision-making, and overall project success rates. Additionally, the integration of AI and data analytics enhances real-time analysis, automation, adaptive practices, and stakeholder relations in project management processes. These advancements contribute to improved project outcomes, increased productivity, and regional development with the construction industry. 3) Execution process group: The analysis reveals valuable insights into the application of various tools, challenges, and impacts associated with the execution process group in PMBOK. These findings shed light on the utilization of tools such as lean techniques, enterprise resource planning (ERP), artificial intelligence (AI), machine learning models, and decision support systems to enhance project execution, productivity, process streamlining, and decision-making in different domains. However, the implementation of these tools is not without challenges. The papers [34], [39] emphasize the challenges of integrating multiple systems, ensuring compatibility, overcoming resistance to change, and addressing complexity and uncertainty, and dealing with technical issues. For example,

are research [40], focuses on the challenges related to preprocessing, algorithm selection, converting models, and conducting case studies in the context of predicting construction labor productivity using ensemble machine learning models. These challenges necessitate comprehensive strategies and skills to implement on successful fully employ the identified tools in project execution. Despite the challenges, the implementation of the identified tools brings about significant impacts on the execution process group. The papers highlight outcomes such as minimizing non-value-added efforts and waste, optimizing resource utilization, reducing project costs, improving project delivery time and quality, enhancing overall productivity and efficiency, and improving decision-making [34], [40], [41], [49]. For instance, a paper [34] discusses the impact of lean techniques, ERP, and AI in construction project management, which results in minimizing waste, optimizing resource utilization, reducing project costs, and improving project delivery time and quality within the construction industry. Furthermore, the papers emphasize the potential of AI and advanced analytics tools in disrupting project management practices. These tools enhance productivity, support decision-making, automate processes, and emphasize stakeholder relationships and risk management [37], [49]. A paper specifically highlights the opening for disruption from AI and advanced analytics tools, underscoring their potential to transform project management practices through increased support, automation, and adaptive practices [49]. In summary, the analysis of the six papers highlights the significance of employing advanced tools and techniques in the execution process group in PMBOK. While the adoption of these tools offers numerous benefits, challenges related to system integration, resistance to change, and technical issues need to be addressed. The positive impacts resulting from their implementation include improved productivity, resource utilization, cost reduction, and enhanced decision-making. Further research and focused efforts are required to overcome the identified challenges and fully harness the potential of these tools for effective project execution.

4) Monitoring and controlling process group: The analysis of the 16 papers reveals significant insights into the utilization of tools, challenges, and impacts related to the monitoring and controlling process group in PMBOK. These papers highlight the application of various advanced tools and techniques such as Artificial Intelligence (AI), Building Information Modeling (BIM), Internet of Things (IoT), Data Mining (DM), machine learning, and image processing algorithms. These tools offer promising capabilities in enabling effective project control and monitoring in the construction domain. They facilitate complex data analysis, performance prediction, bottleneck detection, risk identification, and decision-making support. However, the integration of these tools into project management practices is not without challenges. The papers emphasize the challenges associated with system integration, resistance to change, technology adoption, data accuracy, and project complexity [33], [34], [58]. The integration of multiple systems [45], [54], including voice command, optical character recognition, and LiDAR sensors, as mentioned in the certain paper [34], necessitates the overcoming of compatibility issues and resistance to change. Furthermore, the complexity and uncertainty inherent in construction projects pose additional challenges that demand comprehensive strategies and skills to implement. Despite these challenges, the implementation of the identified tools brings about significant impacts on the monitoring and controlling process group. The papers highlight outcomes such as enhanced efficiency, cost reduction, improved project quality, and

timely delivery [9], [33], [34], [53], [57]. Moreover, the utilization of these tools fosters better decision-making, efficient resource management, improved stakeholder relations, and enhanced risk management [49], [57]. Other paper-based literature review [59] discusses the impact of AI integration in the construction industry, leading to improved job security, reduced accidents, automation of high-risk tasks, and the creation of new job opportunities. In summary, the comprehensive analysis of the 16 papers underscores the significance of employing advanced tools and techniques for the monitoring and controlling process group in (IJACS) International Journal of Advanced Computer Science and Applications Vol. 14, No. 9, 2023 374 | Page www.ijacs.a.thesai.org PMBOK. While the adoption of these tools holds immense potential in project control and monitoring, challenges pertaining to system integration, resistance to change, technology adoption, and project complexity necessitate strategic interventions. However, the positive impacts resulting from their implementation include increased efficiency, cost reduction, improved quality, and informed decision-making, thereby driving effective project management. Further research and focused endeavors are required to address the identified challenges and fully harness the potential of these tools for improved project outcomes.

5) Closing process group: The two papers discussed the application of various tools and technologies in the Closing Process Group of project management according to the PMBOK. First paper [34] highlighted the use of tools like voice command, optical character recognition, and LiDAR sensors for what-if scenarios in construction project management. The challenges identified in this paper included integrating multiple systems, ensuring compatibility, overcoming resistance to change, and addressing complexity and uncertainty in construction projects. The impact of these tools was seen in minimizing non-value-added efforts, optimizing resource utilization, reducing project costs, improving project delivery time and quality, and enhancing overall productivity and efficiency within the construction industry. Second paper [49] focused on the disruptions caused by AI and advanced analytics tools in project management. The tool discussed in this paper were software tools incorporating AI and analytics features. The main challenge highlighted was project managers adapting to these software features. However, the impact of these tools was significant, enhancing project management through increased support, automation, adaptive practices, and emphasis on stakeholder relations and risk management in the closing phase. In summary, both papers emphasized the importance of utilizing advanced tools and technologies in the Closing Process Group of project management. The tools offer solutions to challenges such as system integration, resistance to change, and adaptation to software features. Moreover, the impacts mentioned, such as minimizing waste, optimizing resource utilization, reducing costs, improving quality, and enhancing productivity, align with the desired outcomes of the Closing Process Group in PMBOK.

V. CONCLUSION The findings of this systematic literature review (SLR) provide significant insights into the implementation of AI within the process groups defined in the PMBOK. The results highlight the importance of utilizing advanced tools and techniques in each process group and emphasize the need to address specific challenges for successful full AI implementation. Within the Initiating process group, AI tools and analysis techniques were effective in addressing challenges related to risk assessment, cost prediction, and decision-making. In the Planning process group, the application of AI and data analytics tools improved risk assessment, project selection, cost estimation, reso-

urce allocation, and decision-making. The Execution process group experienced improved productivity, resource utilization, cost reduction, and enhanced decision-making through the use of advanced tools and techniques. The Monitoring and Controlling process group also demonstrated the potential of advanced tools and techniques in achieving increased efficiency, cost reduction, improved quality, and informed decision-making. Lastly, the Closing process group highlighted the value of incorporating AI, lean techniques, ERP, and machine learning, resulting in benefits such as cost reduction, efficiency improvement, and project closure success. Overall, this SLR provides valuable insights for organizations aiming to integrate AI into their project management practices. By aligning the challenges and impacts with the PMBOK process groups, this research offers a structured framework that allows organizations to navigate the implementation of AI effectively. Understanding the specific challenges and impacts associated with each process group can help organizations address these issues and capitalize on the potential benefits of AI in project management, ultimately leading to improved project outcomes and increased productivity.

A. Limitations of Study The limitation of this study is that it should be noted that the authors have not explored or presented specific solutions to address the identified challenges discussed in the literature.

B. Future Works In future research, it is important to conduct a bibliographic analysis to identify relevant tools and map them into a specific framework. Additionally, creating a comprehensive mapping of the identified challenges within the framework will provide a better understanding of their origins and facilitate the search for effective solutions.

REFERENCES [1] Intelligence Enabled Project Management: A Systematic Literature 10.3390/app13085014. [2] Technol, vol. 100, no. 16, pp. 50005012, 2022, doi:10.5281/zenodo.7134073. [3] to the Impact of International Conference of Women in Data Science at Taif University (WiDSTaif), 2021, doi:10.1109/WiDSTaif52235.2021.9430234. [4] S. Bento, L. Pereira, R. Gonçalves, D. Dias, and R. L. da Costa, vol. 13, no. 2, pp. 143163, 2022, doi:10.1504/ijtip.2022.126841. [5] C. Engel, P. Ebel, and B. - Effect Relationship of AI Characteristics, Project Management Systems and Organisation, vol. 47, no. February, pp. 166181, 2021, doi:10.1007/978-3-030-86797-3_12. [6] C. Bodea, D. Ronggui, O. Stanciu, and C. Mitea, Artificial Intelligence impact in Project Management. International project management association, 2020. (IJACSA) International Journal of Advanced Computer Science and Applications Vol. 14, No. 9, 2023 375 | Page www.ijacsa.thesai.org [7] int39, 2022. [8] A. Belharet, U. Bharathan, B. Dzingina, N. Madhavan, C. Mathur, and Y. - Manage 53, 2020. [9] Generation Computer Systems, vol. 123, pp. 8593, 2021, doi:10.1016/j.future.2021.04.013. [10] A.T.T Association of Cost Engineers, vol. 30, no. 6. The Association, 24. doi:info:doi/. [11] -driven project buffersizing in 2022, doi:10.1016/j.autcon.2022.104134. [12] P.M.I. (ASV), The standard for project management and a guide to the project management body of knowledge (PMBOK) 7th Edition., no. July. 2021. [13] PMI, A Guide to the Project Management Body of Knowledge (PMBOK) Fifth Edition, vol. 6. Project Management Institute, Inc., 2013. [14] <https://bdtechtalks.com/2020/04/09/what-is-narrow-artificial-intelligence-ani/> (accessed Jun. 12, 2023). [15] DAMA International, DAMA-knowledge 2nd Edition. 2017. [16] H.K. Mohamed, S.M. El-Debeiky, H.M. Mahmoud, and K.M. El465. doi:10.1109/ICCES.2006.320491. [17] R. Shukla, P. Sharm MINING - A Study of Web data

pattern detecting methodologies and its 10.1109/IDEA.49133.2020.9170690. [18] Learning Alg 10.1088/1742-6596/1168/3/032132. [19] Evaluation and Application of a Project Charter Template to Improve Management Journal, vol. 31, no. 1, pp. 1423, Mar. 2000, doi:10.1177/875697280003100104. [20] 20, 2018, [Online]. Available: www.ijarse.com [21] C. and P. M.) Amponsah, Richard (Sch Project Management Practice in Ghana with Focus on Agriculture, vol. 8, no. July, p. 441, 2010, [Online]. Available: [https://researchbank.rmit.edu.au/view/rmit:10389\[22\]-2535611/v1](https://researchbank.rmit.edu.au/view/rmit:10389[22]-2535611/v1). [23] search Circular, no. E-C168, pp. 68, 2012. [24] 24, 1987, doi: DOI:10.1017/S0890060400000111. [25] B. A. Kitchenham and S. Charters Systematic Literature Reviews in Software Engineering (Software Technical Report EBSE2007-001. Keele University and Durham University Joint Report, no. January, 2007. [26] International Conference Proceeding Series, pp. 123129, 2020, doi:10.1145/3378936.3378949. [27] P. Marnada, T. Raharjo, B. Hardi a management challenge in handling scope and change: A systematic 300, 2021, doi:10.1016/j.procs.2021.12.143. [28] -data mining integrated digital twin no. July 2020, p. 103564, 2021, doi:10.1016/j.autcon.2021.103564. [29] FAC-Papers On Line, vol. 55, no. 10, pp. 32863291, 2022, doi:10.1016/j.ifacol.2022.10.127. [30] S. M. Albogami, M. K. A. B. M. Ariffin, K. A. Ahmad, and E. E. B. shafar the theory of evidence method for project risk assessment 24, 2021, doi:10.3390/math9243225. [31] Building Engineering, vol. 35, no. October 2020, 2021, doi:10.1016/j.job.2020.102001. [32] F. Sanchez, E. Bonjour, J. -Based on Bayesian Network for Improving Project Management Maturity: An Application to Reduce Cost Overrun Risks in Engineering Comput Ind, vol. 119, p. 103227, 2020, doi: <https://doi.org/10.1016/j.compind.2020.103227>. [33] 10.1155/2021/3585878. [34] V. Vickrle ant techniques, enterprise resource planning and artificial intelligence Technology and Engineering, vol. 7, no. 6C2, pp. 147153, 2019, [Online]. Available: <https://www.scopus.com/inward/record.uri?eid=2-s2.0-85067860685&partnerID=40&md5=b12e982df42fa911504460e636f9f6a2> [35] construction projects using Machine learning and image processing 563, 2022, doi:10.1016/j.matpr.2022.03.137. [36] -learning and EDM to monitor and predict a complex project with a GERT-type network: A multi-po 109256, 2023, doi:10.1016/j.cie.2023.109256. [37] of Recent Technology and Engineering, vol. 8, no. 2 Special Issue 3, pp. 13731377, 2019, doi:10.35940/ijrte.B1256.0782S319. [38] 105688, 2020, doi: <https://doi.org/10.1016/j.dib.2020.105688>. [39] I. T. Afolabi, A. A. Adebisi, E. G. Chukwurah, and C. P. Igboke, Notes in Engineering and Computer Science, 2019, pp. 233237. [Online]. Available: <https://www.scopus.com/inward/record.uri?eid=2-s2.0-85083396939&partnerID=40&md5=dd33acf7d926df5a30ff59cfe97760ed> [40] I. Karatas and A. Budak -ensemble machine learning model in predicting construction labor Management, vol. ahead-of-p, no. ahead-of-print, Jan. 2022, doi:10.1108/ECAM-08-2021-0692. [41] Information Technologies, vol. 20, no. 2, pp. 7992, 2020, doi:10.2478/cait-2020-0018. [42] M. Relichandl. Ni-Papers On Line, vol. 54, no. 1, pp. 10921097, 2021, doi:10.1016/j.ifacol.2021.08.128. (IJACSA) International Journal of Advanced Computer Science and Applications Vol. 14, No. 9, 2023 376 | Page www.ijacsa.thesai.org [43] udget in-Papers On Line, vol. 55, no. 10, pp. 32383243, 2022, doi:10.1016/j.ifacol.202

2.10.140.[44]Ind Eng, vol.180, no. April, 2023, doi:10.1016/j.cie.2023.109261.[45]M.M.A.Ranesh, S.J.Samuel, R.Natchadalingam, and P.Jeyanthi, Neural Network and Balance Score card to Improve the Success Rate of Communication and Aerospace Technology, 2022, pp.12161221. doi:10.1109/ICECA55336.2022.10009299.[46]O.Allal-Chérif, V.Simon-purchasing: How artificial intelligence can redefine the purchasing 76, 2021, doi:10.1016/j.jbusres.2020.11.050.[47]data-driven fr doi:10.1038/s41598-022-19728-x.[48]pp.68776889, 2023, doi:10.1007/s00500-023-08133-9.[49]6, pp.15701599, Jan.2021, doi:10.1108/ITP-09-2020-0639.[50]R.Hassani and Y.El Bouzekri El International Journal of Advanced Trends in Computer Science and Engineering, vol.8, no.6, pp.33963404, 2019, doi:10.30534/ijatcse/2019/114862019.[51]M.J.Bianchi, E.C.Conforto, E.Rebentisch, D.C.Amaral, S.O. vol.69, no.6, pp.35583571, 2022, doi:10.1109/TEM.2021.3101179.[52]analysis for construction project: Combined cascading effect and https://doi.org/10.1016/j.ssci.2021.105410.[53]search-optimized deep learning technique for mapping construction cash 138, 2020, doi:10.1016/j.chaos.2020.109869.[54]Intell Humaniz Comput, 2021, doi:10.1007/s12652-020-02668-7.[55]rship orientation vol.226, no. April 2019, p.107621, 2020, doi:10.1016/j.ijpe.2020.107621.[56]construction projects site Built Environment Project and Asset Management, vol.12, no.5, pp.738753, 2022, doi:10.1108/BEPAM-03-2021-0047.[57]Improving the Prediction 22, 2022, doi:10.3390/math10224189.[58](Stuttg), p.170460, 2022, doi:10.1016/j.ijleo.2022.170460.[59]intelligence in South African construction project management Engineering and Operations Management, 2021, pp.148162.[Online]. Available: <https://www.scopus.com/inward/record.uri?eid=2-s2.0-85121110767&partnerID=40&md5=317512e166c09d24a3f00e4eefa0645a>